moons of Uranus, looks disconcertingly jumbled and may have had such an origin.

The American planetary geologist Eugene Shoemaker proposes that many moons in the outer Solar System have been annihilated and re-formed—not just once but several times each over the 4.5 billion years since the Sun and the planets condensed out of interstellar gas and dust. The picture emerging from the *Voyager* reconnaissance of the outer Solar System is of worlds whose placid and lonely vigils are spasmodically troubled by interlopers from space; of world-shattering collisions; and of moons re-forming from debris, reconstituting themselves like phoenixes from their own ashes.

But a moon that lives very close to a planet cannot re-form if it is pulverized—the gravitational tides of the nearby planet prevent it. The resulting debris, once spread out into a ring system, might be very long-lived—at least by the standard of a human lifetime. Perhaps many of the small, inconspicuous moons now orbiting the giant planets will one day blossom forth into vast and lovely rings.

These ideas are supported by the appearance of a number of satellites in the Solar System. Phobos, the inner moon of Mars, has a large crater named Stickney; Mimas, an inner moon of Saturn, has a big one named Herschel. These craters—like those on our own Moon and, indeed, throughout the Solar System—are produced by collisions. An interloper smashes into a bigger world and makes an immense explosion at the point of impact. A bowl-shaped crater is excavated, and the smaller impacting object is destroyed. If the interlopers that dug out the Stickney and Herschel craters had been only a little larger, they would have had enough energy to blow Phobos and Mimas to bits. These moons barely escaped the cosmic wrecking ball. Many others did not.

Every time a world is smashed into, there's one less interloper—something like a demolition derby on the scale of the Solar System, a war of attrition. The very fact that many such collisions have occurred means that the rogue worldlets have been largely used up. Those on circular trajectories around the Sun, those that don't intersect the orbits of other worlds, will be unlikely to smash into a